HOSAKOTA VENKATA NAGA THARUN

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OBJECTIVE

AI Engineer with 2.5 years of experience designing and building production AI applications powered by LLMs, RAG, and agentic workflows on Azure. Skilled in multi-agent orchestration with LangGraph, hybrid retrieval pipelines (FAISS and BM25 with cross-encoder reranking), prompt engineering, and serverless AI backends built on Azure OpenAI and Azure AI Foundry Agents. Focused on reliable AI systems built with validation, error handling, monitoring, and root-cause analysis so failures do not recur.

EXPERIENCE

Mouri Tech Pvt Ltd **March 2024 - Present**
Applied AI Engineer

Financial Statement Analyzer

- Built a serverless Python backend on Azure Functions with Azure AI Foundry Agents (GPT-4.1, code interpreter) to extract structured summaries from financial statement images; a representative 42-image batch takes ~2 min 50 sec
- Root-caused API rate-limit failures that occurred consistently past 15-20 images per batch under bulk base64 image load, then implemented corrective actions to prevent recurrence: odd/even index routing across two agents, retry logic, and automatic failover between agents; zero rate-limit failures observed since
- Designed a 3-phase prompt pipeline (extraction, verification, summary) with a verification phase enforcing strict data-fidelity rules to validate extracted financial data and limit hallucination
- Monitored runs through Azure Application Insights logs across monthly and yearly cycles to confirm fixes held
- Secured the service with Azure Identity (DefaultAzureCredential) for passwordless, managed-identity authentication

Multi-Agent AI Assistant Platform

- Designed a multi-agent system with LangGraph StateGraph orchestration, Azure OpenAI (GPT-4.1), and FastAPI; a context-aware router dispatches queries across 6 specialized agents (calculator, weather, HR policy, search, reminder, smalltalk)
- Built a 3-stage hybrid RAG pipeline combining dense FAISS search (70% weight) and sparse BM25 keyword search (30% weight) with cross-encoder reranking, grounding responses across 30+ internal policy documents
- Implemented priority-based short-circuit routing to cut unnecessary LLM calls
- Built a Human-in-the-Loop approval workflow for the reminder agent: collects the recipient email, message, and reminder time over multiple turns, then asks for confirmation ("Are you sure?") before the reminder email is sent
- Implemented follow-up handling across all agents: retrieves the tool used in the previous turn and decides whether a new request relates to it or needs up to 10 past conversation turns, and re-asks for approval before resending a reminder
- Added real-time SSE streaming of graph execution and a dashboard tracking token usage, latency, and agent distribution

Workato AI Agents - LeadPilot AI and Outlook Automation

- Built LeadPilot AI (Workato Genie), which analyzes HubSpot contact details and Salesforce matching context to score leads from 0-100, assign High, Medium, or Low priority, and recommend the next sales action, owner role, and follow-up task
- Designed LeadPilot to distinguish exact email matches from company-only matches so same-company contacts are not treated as one person, returning structured recommendations for the Workato recipe to act on
- Built an Outlook automation agent (Workato Genie) that summarizes emails, sends new emails and replies, and suggests next actions, using Workato recipes as tools: get unread, get all, get details, mark as read, send, and delete
- Added a human-approval step ("Are you sure?") before the agent deletes any email

TECHNICAL SKILLS

- **Programming Languages**: Python
- **Generative AI**: LLMs, Agentic AI, AI Agents, Multi-Agent Systems, Prompt Engineering, RAG, Hybrid Search, NLP
- **Agent & LLM Frameworks**: LangGraph, LangChain, Semantic Kernel, Google ADK
- **Reliability & Debugging**: Validation, Error Handling, Retry and Failover, Logging, Monitoring, Root-Cause Analysis
- **Vector Search**: FAISS, BM25, Cross-Encoder Reranking, Vector Databases
- **Cloud**: Azure OpenAI, Azure AI Foundry Agents, Azure Functions, Azure Identity/AD, Application Insights, Databricks
- **Databases**: PostgreSQL, Cosmos DB
- **Web Frameworks & APIs**: FastAPI, REST APIs
- **Automation**: Workato, Playwright, GitHub Actions (CI/CD)
- **Tools and Platforms**: Git, GitHub, GitLab, Docker, Postman, VS Code, GitHub Copilot, Cursor

EDUCATION

Jawaharlal Nehru Technological University Anantapur (JNTUA) **2019 - 2023**
Bachelor of Technology, Electronics and Communication Engineering

- **Percentage:** 74%